



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

Prof Masoumeh Kazemi



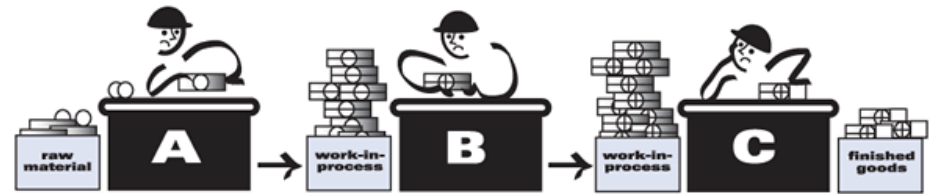
Agenda

- Chapter 15 Probabilistic Models – Queuing Theory

Queuing theory

- Mathematical approach to the analysis of waiting lines

- Emergency rooms/Walk-in clinics
- Bank branches
- Call/service centers
- Telecommunication networks
- Production lines

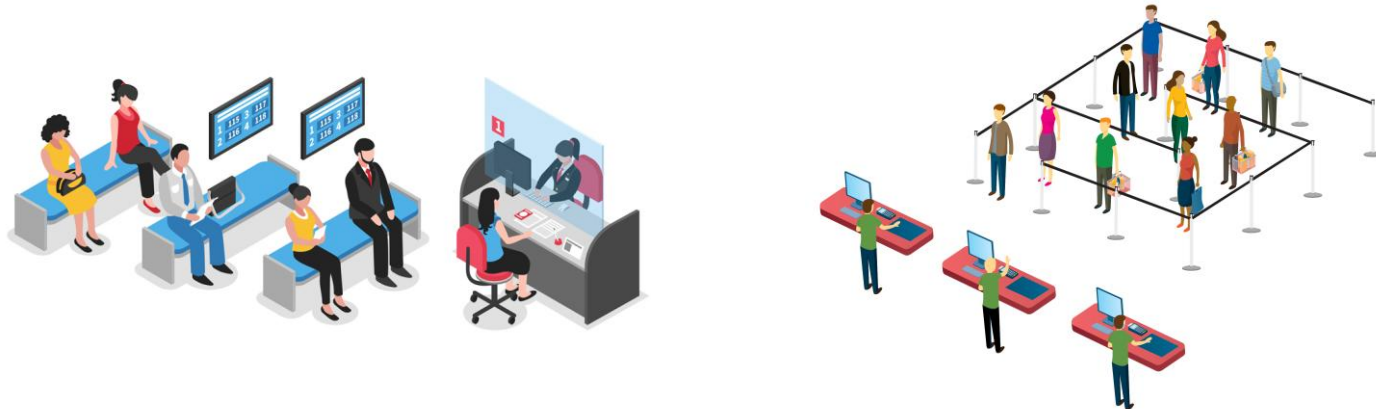


- Goal of queuing analysis is to minimize the sum of two costs
 - Customer waiting costs
 - Service capacity costs

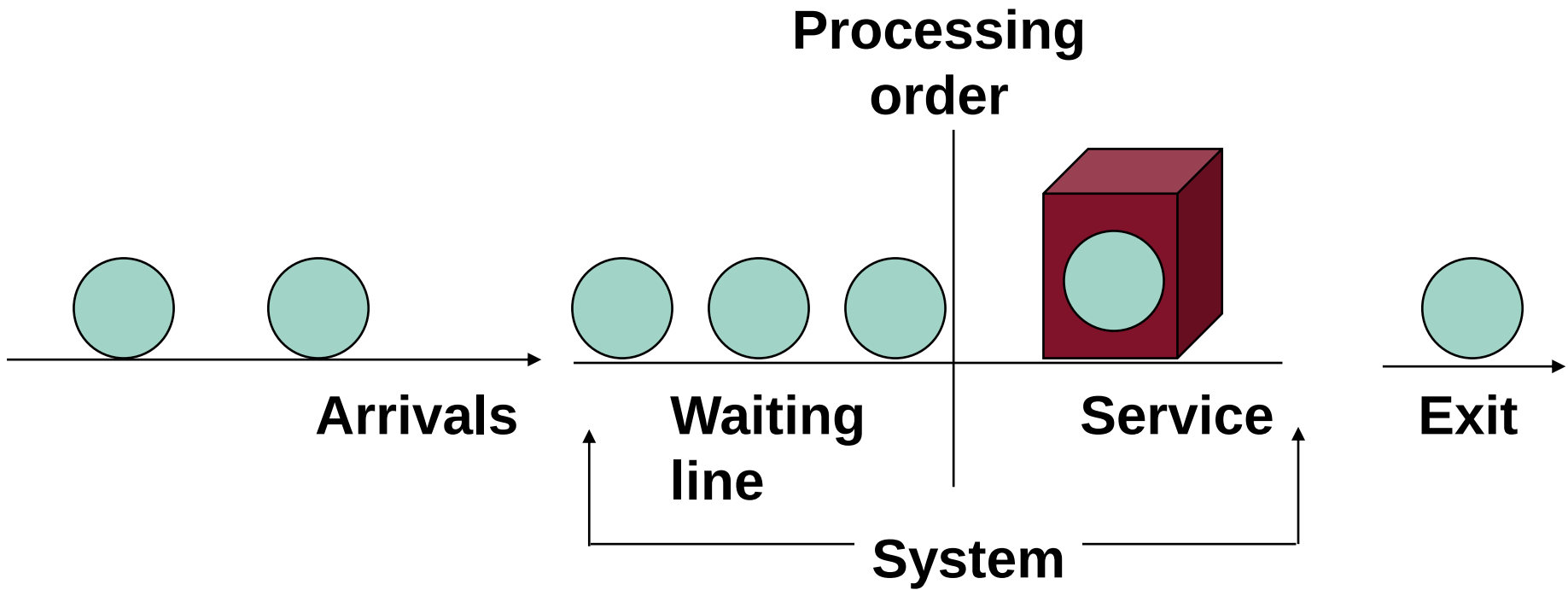


System Characteristics

- Population Source
 - Infinite source: customer arrivals are unrestricted
 - Finite source: number of potential customers is limited
- Number of servers (channels)
- Arrival and service patterns
- Queue discipline (order of service)

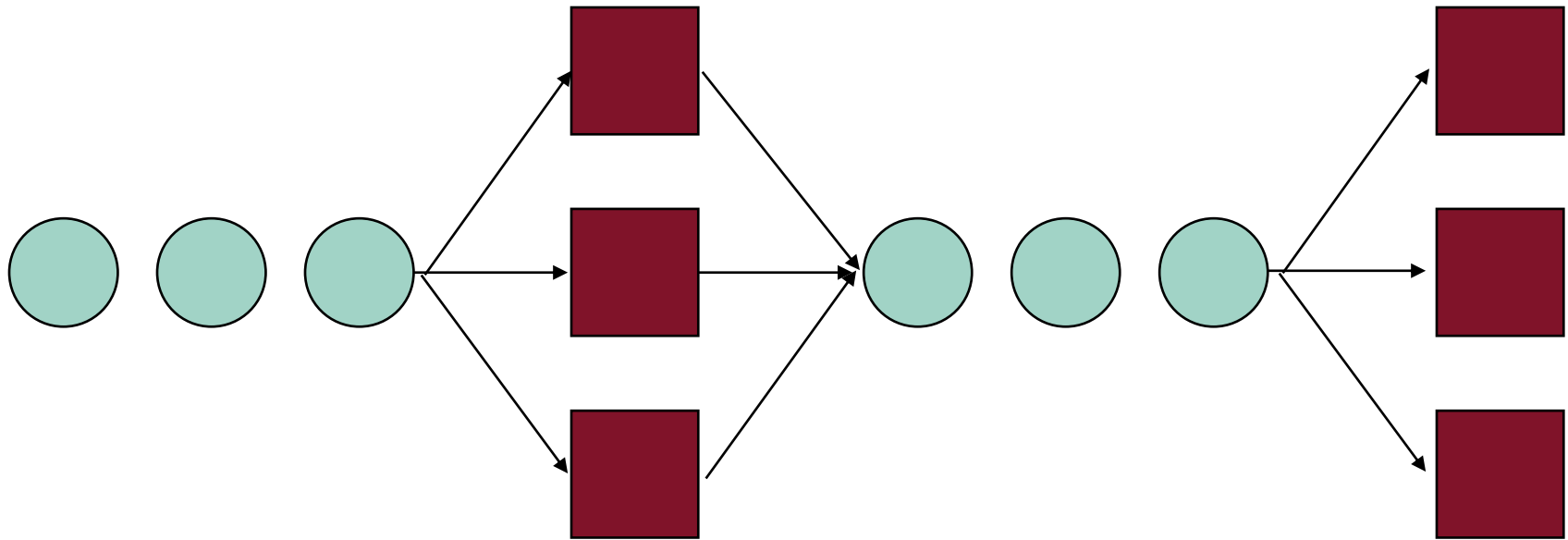


Elements of Queuing Systems



Queuing Systems

Multiple servers (channels)

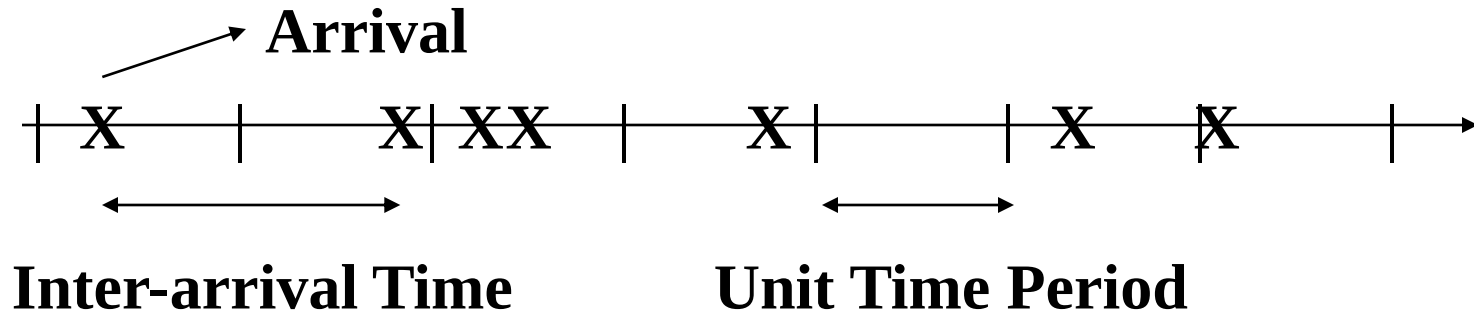


Multiple stages

Examples of Queue Structures

	Single Phase	Multiphase
Single Channel	One-person barber shop	Car wash
Multi-channel	Bank tellers' windows	Hospital admissions

Modeling the Arrivals



Inter-arrival Time = Time between two consecutive arrivals (a random quantity)

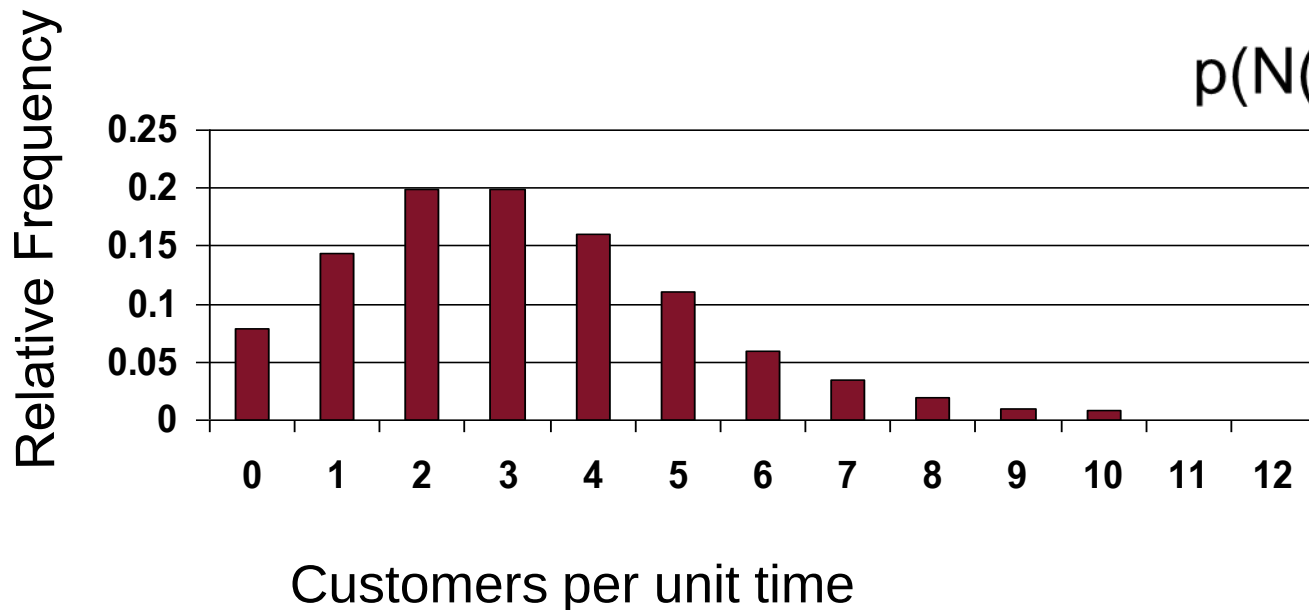
Number = Number of arrivals in a unit time period (also a random quantity)

How to represent this randomness ?

Use common probability distributions

Arrival Rate: Poisson Distribution

- Discrete probability distribution
 - The probability of a given number of events occurring in a fixed interval of time at a known constant mean rate (λ), independently of the time since the last event



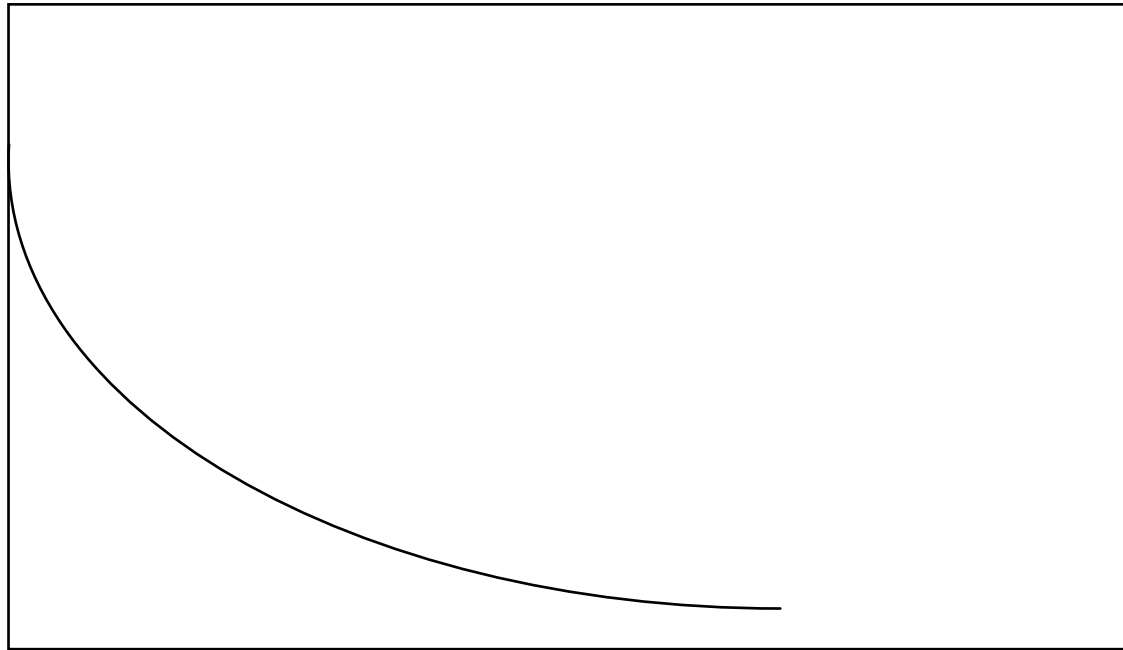
$$p(N(t)=n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

- Exponential distribution models the time between these events.

Service Time: Exponential Distribution

- Continuous probability distribution
 - Models the time elapsed between events in a Poisson process
 - Average processing time ($1/\mu$) $\rightarrow$ Average service rate (μ)

Relative Frequency (%)



$$f(t) = \mu e^{-\mu t}$$

Time

System Performance at steady state

- Average number of customers waiting (in the line or in the system)
- Average waiting time of customers (in the line or in the system)
- Probability that an arrival will have to wait
- Cost

Basic Queuing Models-Infinite Source

- Identify values of the following parameters:
 - λ = arrival rate,
 - μ = service rate
 - M = number of channels or servers
- Use the above information to calculate:
 - ρ = utilization (percentage of the time the servers are busy per shift)
 - N_q = Average number of customers in queue
 - N_s = Average number of customers in system
 - T_q = Average Waiting time in queue
 - T_s = Average Waiting time in system
 - $P(n)$ = Probability of n customers in system

Steady State Results for Single Service Channel System with exponential (Poisson) arrival and exponential service time (M/M/1 queue)

- Equilibrium: the rate of customers entering the system equals the rate of customers leaving it.
- Utilization (ρ) must be < 1 $\rho = \lambda / \mu$
- Relationship between expected number of customers in the system and waiting times (Little's Law)

$$N_s = \lambda T_s$$

$$N_q = \lambda T_q$$

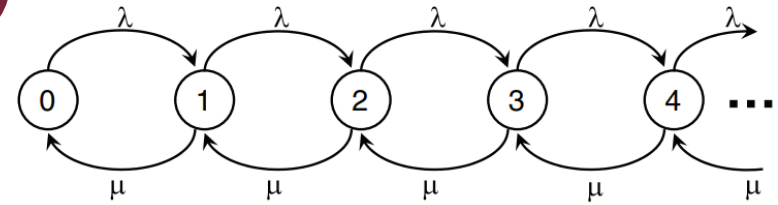
- Relationship between waiting time in the system and waiting time in the queue

$$T_s = T_q + 1 / \mu$$

- Average number of customers in the system (N_s) is problem dependent

Model: Single-Service Channel, exponential distribution of arrival and service times (M/M/1)

- Markov chains



Number in System

- Probability of no customer in the system

$$P(0) = 1 - \lambda / \mu = 1 - \rho$$

- Probability of exactly n customers in the system

$$P(n) = (1 - \rho) \rho^n$$

- The expected number of customers in the system

$$N_s = \rho / (1 - \rho) = \lambda / (\mu - \lambda)$$

- Other service measures can be obtained from the basic relationship formulas.

Other Measures Derived

$$T_s = N_s / \lambda = 1 / (\mu - \lambda)$$

$$T_q = T_s - (1 / \mu) = \lambda / \mu (\mu - \lambda)$$

$$N_q = \lambda T_q = \lambda^2 / \mu (\mu - \lambda)$$

Steady State Results for Single Service Channel System with exponential (Poisson) arrival and exponential service time

- Utilization (should be <1): $\rho = \lambda / \mu$
- Average Number in Line (N_q): $N_q = \lambda^2 / \mu (\mu - \lambda)$
- Average Number in System: $N_s = \lambda / (\mu - \lambda)$
- Average Time for Customers:
 - → Wait in Line: $T_q = \lambda / \mu (\mu - \lambda)$
 - → Wait in System: $T_s = 1 / (\mu - \lambda)$

Example

Drive-up window at a fast food restaurant.

Customers arrive at the rate of 25 per hour.

The employee can serve one customer every two minutes.

Assume Poisson arrival and exponential service rates.

- A) What is the average utilization of the employee?
- B) What is the average number of customers in line?
- C) What is the average number of customers in the system?
- D) What is the average waiting time in line?
- E) What is the average waiting time in the system?
- F) What is the probability that exactly two cars will be in the system?
- G) What is the probability that there will be more than 5 cars waiting in the system?

Solution

A) What is the average utilization of the employee?

$$\lambda = 25 \text{ cust / hr}$$

$$\mu = \frac{1 \text{ customer}}{2 \text{ mins (1hr / 60 mins)}} = 30 \text{ cust / hr}$$

$$\rho = \frac{\lambda}{\mu} = \frac{25 \text{ cust / hr}}{30 \text{ cust / hr}} = \mathbf{.8333}$$

Solution

B) What is the average number of customers in line?

$$N_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(25)^2}{30(30-25)} = \mathbf{4.167}$$

C) What is the average number of customers in the system?

$$N_s = \frac{\lambda}{\mu - \lambda} = \frac{25}{(30-25)} = \mathbf{5}$$

Solution

D) What is the average waiting time in line?

$$T_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{25}{30(30-25)} = .1667 \text{ hrs} = \mathbf{10 \text{ mins}}$$

E) What is the average waiting time in the system?

$$T_s = \frac{1}{\mu - \lambda} = \frac{1}{30-25} = .2 \text{ hrs} = \mathbf{12 \text{ mins}}$$

Solution

F) What is the probability that exactly two cars will be in the system (one being served and the other waiting in line)?

$$p_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n$$

$$p_2 = \left(1 - \frac{25}{30}\right) \left(\frac{25}{30}\right)^2 = \mathbf{.1157}$$

Solution

G) What is the probability that there will be more than 5 cars waiting in the system?

$$p_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n$$

$$\begin{aligned} P(n > 5) &= 1 - (p(0) + p(1) + p(2) + p(3) + p(4) + p(5)) \\ &= 1 - (0.166 + 0.138 + 0.115 + 0.096 + 0.08 + 0.066) = \\ &= 0.339 \end{aligned}$$

General features of queuing systems with M servers (channels)

- For steady state to exist:
 - If population is infinite,
 - utilization $\rho = \lambda / M\mu$ must be strictly less than 1.0 if arrival or service process is probabilistic
 - utilization $\rho = \lambda / M\mu$ can be equal to 1.0 only if both arrival and service processes are deterministic
 - If population is finite,
 - utilization $\rho = \lambda / M\mu$ can be any positive value if arrival or service process is probabilistic

Probabilistic Models and Computer Simulation

- Numerical models with probabilistic information
- Analytical results for simple situations
- Simulation study or numerical experiments for “realistic” analysis
- May not be aiming at “optimal solutions”

Computer simulation

- Widely used in industry
- Less structured or more complex queuing systems or networks
- More general probability distributions other than exponential/Poisson (Markovian) for arrival or service
- Basic queuing models can be used to verify simulation models
- Numerical experimentation

Computer simulation

- Answers “ what-if ” questions
- Work with other models such as inventory, production planning, facility layout, material handling, warehousing, CAD/CAM and optimization
- Use animation to visualize the operations
- Must be performed properly
- Can be quite expensive and time-consuming